



Computer Ethics and Social Issues (CSEg5206)

Crhr: 3 Department: CSE

Program: Undergraduate Regular

Batch: 5th Year

Instructor: Teklu Urgessa (PhD, Associate Professor)

Course Outline

Introduction to Ethics:

- *Ethical Theories,*
- *Comparing Workable Ethical Theories,*
- *Morality of Breaking the Law*

Networked Communications:

- *Spam,*
- *Censorship,*
- *Filtering,*

Freedom of Expression, Intellectual Property and Constitutional Issues:

- *Intellectual Property Rights*
- *Protecting Intellectual Property,*
- *Fair Use,*
- *New Restrictions on Use*
- *Legitimacy of Intellectual Property*

Protection for Software Information Privacy:

- *Perspectives on Privacy,*
- *Information Disclosures,*
- *Data Mining,*
- *Examples of Consumer Backlash.*

Professional Ethics and Responsibilities:

- *How Well Developed Are the Computing Professions?*
- *Software Engineering Code of Ethic*

Course Delivery

- *Lectures*
- *Assignment and presentation*
- *Active class participation*
- *Reading assignments*

Assessment

- *Assignment and Presentation* 15
- *Seminar* 10
- *Class Activity* 10
- *Mid exam* 25
- *Final exam* 40

Related Research Areas

- *Ethics and Policies related ICT*
- *Internet Attacks*
- *Prevention of Attacks (cyber-security)*
- *Privacy and Security issues in the digital age*
- *AI Ethics*

References

1. *Michael J. Quinn. Ethics for the information Age, 2nd edition*
Pearson Addison Wesley, 2006.
2. *Rahul Tongia, Eswaran Subrahmanian, and V.S.*
Arunachalam. Information and Communications
Technology for Sustainable Development: Defining a
lobal Research Agenda. Allied PUblishers, 2005.

Computer Ethics and Social Issues (CSE5206)

Introduction to Ethics

- ☐ Introduction
- ☐ Ethical Theories
- ☐ Comparing Workable Ethical Theories
- ☐ Morality of Breaking the Law

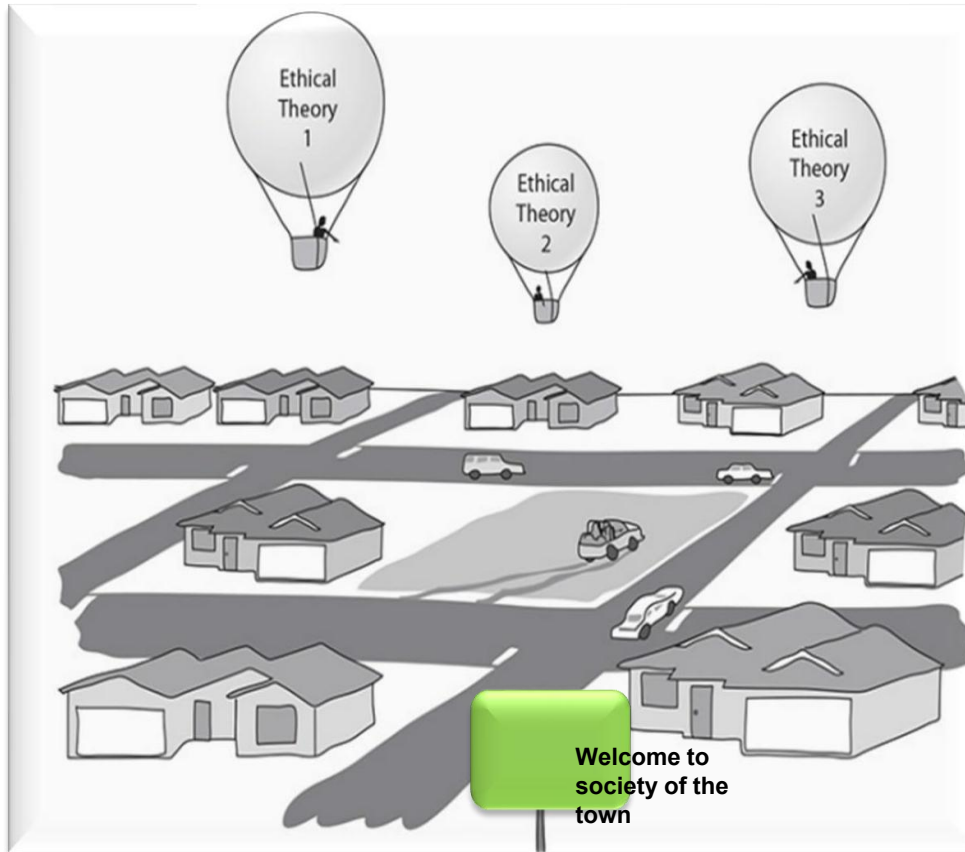
The Ethical Point of View

- Virtually everyone shares “core values”, desiring:
 - Life
 - Happiness
 - Ability to accomplish goals
- Two ways to view world
 - **Selfish point of view**: consider only own self and its core values
 - **Ethical point of view**: respect other people and their core values

Defining Terms

- **Society**
 - Association of people organized under a system of rules to advance goods to its members over time
 - Rules: *Obligations and prohibition* on members to advance the good of society's members over time
- **Morality**
 - A society's *rules of conduct*
 - What people *ought /ought not* to do in various situations
- **Ethics**
 - is the *philosophical study* of morality,
 - a *rational examination* into people's moral beliefs and behaviour
 - evaluation of people's behavior
- **Philosophy:** the study of the *fundamental nature* of:
 - knowledge
 - Reality
 - Existence

Analogy of Important Terminologies



☐ **Society**

is like a town full of people driving cars

☐ **Morality**

is the road network within the town

☐ **People**

ought to keep their cars on the roads/morality

☐ People *doing ethics* (observers: Judges) are in balloons floating above the town

☐ Staying on the roads: acting morally or

☐ **Taking shortcuts:** acting immorally

Why Study Ethics?

- **Ethics:** a way to decide the best thing to do
- New problems accompany *new technologies*
- “Common wisdom” *may not exist* for novel situations brought about by new technologies
- Take how *smartphones, AI, IoTs* and *Social media apps* have changed how
 - We spend our time and
 - Interact with others
- New *products and services* have brought many common benefits to society as well as
- It may also *selfishly exploited* for personal gain, even if that *reduces their overall benefit* for the rest of the society

More on Ethics

- Ethics: rational, systematic analysis
 - “Doing ethics”: answers need explanations
 - Explanations: facts, shared values, logic
- Ethics: voluntary, moral choices
- Workable ethical theory: produces explanations that might be persuasive to a skeptical, yet open-minded audience

Good Ethical Theory Supports

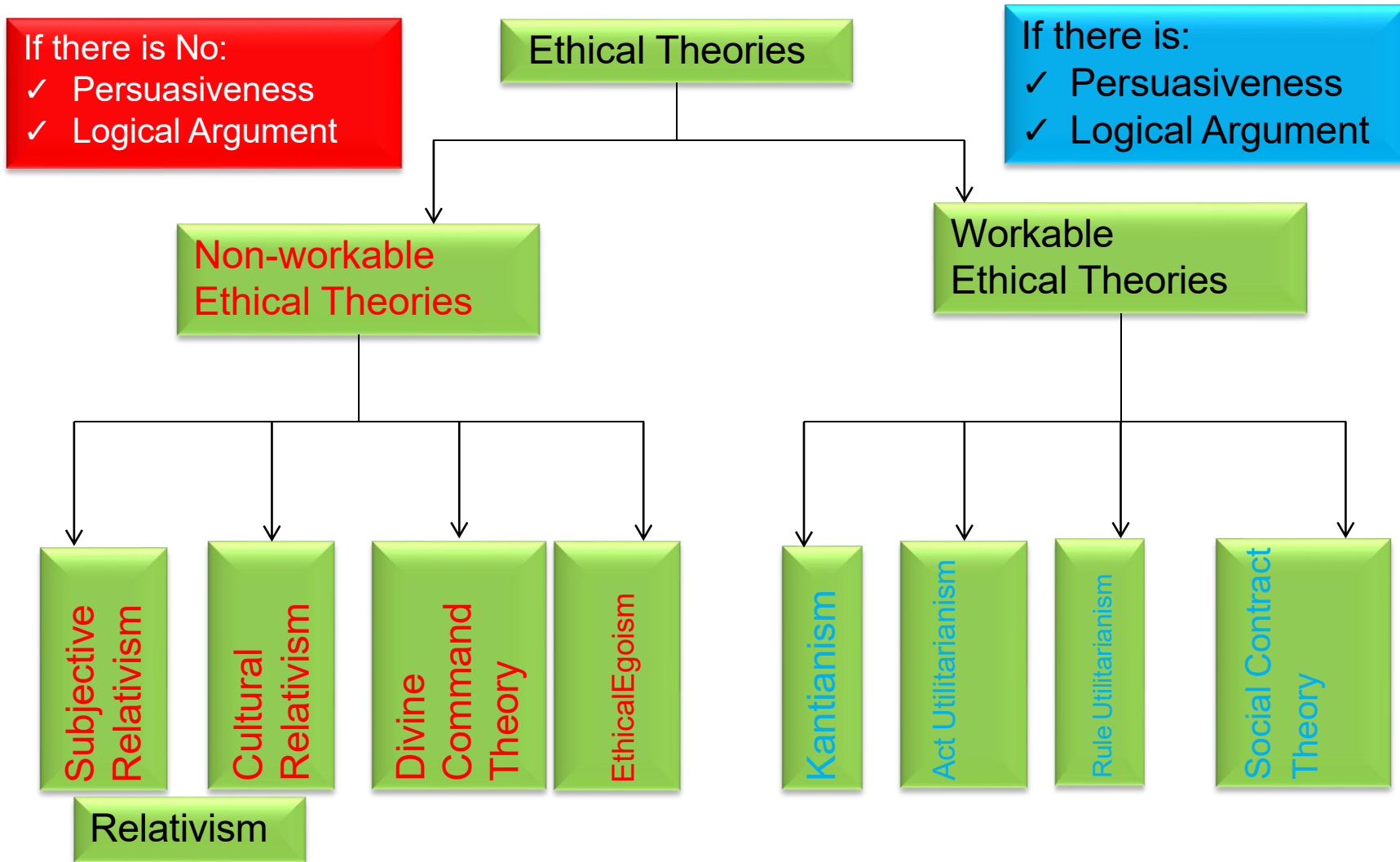
- Persuasiveness
- Logical Arguments



Types Ethical Theories

- ☐ Non Workable Ethical Theories
- ☐ Workable Ethical Theories

Categorization of Ethical Theories



Subjective Relativism

(subject to tolerance and What someone wants to do)

What Is Relativism?

- Relativism
 - No universal norms of right and wrong
 - One person can say “X is right,” another can say “X is wrong,” and
 - both can be right
- Subjective relativism
 - Each person decides right and wrong for himself or herself
 - “What’s right for you may not be right for me”

Case for Subjective Relativism

- Well-meaning and intelligent people disagree on moral issues
- Ethical debates are disagreeable and pointless

Case Against Subjective Relativism

- Blurs distinction between doing what you think is right and doing what you want to do
- Makes no moral distinction between the actions of different people
- SR and tolerance are two different things
- Decisions may not be based on reason
- **Not a workable ethical theory**

Cultural Relativism

***(subject to place and
Time)***

Cultural Relativism in a Nutshell

- What is “right” and “wrong” depends upon a society’s actual moral guidelines
- These guidelines vary from place to place and from time to time
- A particular action may be right in one society at one time and wrong in other society or at another time

Case for Cultural Relativism

- Different social contexts demand different moral guidelines
- It is arrogant for one society to judge another

Case Against Cultural Relativism

- Because two societies **do** have different moral views doesn't mean they **ought to** have different views
- It doesn't explain how moral guidelines are determined
- What if there are no cultural norms?
- It doesn't account for evolution of moral guidelines.
- It provides no way out for cultures in conflict
- Existence of many acceptable practices does not imply all practices are acceptable (many/any fallacy)
- Societies do, in fact, share certain core values
- Only indirectly based on reason
- Not a workable ethical theory

Divine Command Theory

***(Absolute Authority → “God ’ s will
is good ”)***

Overview of Divine Command Theory

- Good actions: those aligned with God's will
- Bad actions: those contrary to God's will
- Holy books reveal God's will
- We should follow holy books as moral decision-making guides

Divine Command Theory in Action



Case **for** Divine Command Theory

- We owe obedience to our Creator
- God is all-good and all-knowing
- God is the ultimate authority

Case **Against** Divine Command Theory

- Different holy books disagree
- Society is multicultural, secular
- Some modern moral problems not addressed in scripture e.g. Internet
- Based on *obedience*, *not reason*
- Not a workable ethical theory

Ethical Egoism

(selfish point of View)

Definition of Ethical Egoism

- Each person should focus exclusively on his or her self-interest
- Morally right action: action that provides self with maximum long-term benefit

Case **for** Ethical Egoism

- It is practical since we are already inclined to do what's best for ourselves
- It's better to let other people take care of themselves
- The community can benefit when individuals put their well-being first
- Other moral principles are rooted in the principle of self-interest

Case Against Ethical Egoism

- An easy moral philosophy may not be the best moral philosophy
- We know a lot about what is good for someone else
- Self-interest can lead to blatantly immoral behavior
- Other moral principles are superior to principle of self-interest
- People who take the good of others into account lead happier lives
- By definition, does not respect the ethical point of view
- Not a workable ethical theory

Case *Against* Ethical Egoism

- An easy moral philosophy may not be the best moral philosophy
- We do, in fact, know a lot about what is good for someone else
- A self-interested focus can lead to blatantly immoral behaviour.
- Other moral principles are superior to the principle of self-interest
- People who take the good of others into account live happier lives.

Kantianism
***(Categorical Imperative 1 &
2)***

Critical Importance of *Good Will*

- Good will: the **desire** to do the right thing
- Immanuel Kant: Only thing in the world that is good without qualification is a good will
- **Reason** should cultivate desire to do right thing
- Good without qualification: a good will is always good, and does not require anything else to be good

Categorical Imperative (1st Formulation)

Act only from *moral rules* that you can at the same time will to be *universal moral laws*.

Categorical Imperative: is a way of evaluating motivations for action

Illustration of 1st Formulation

- Question: Can a person in dire straits make a promise with the intention of breaking it later?
- Proposed rule: “I may make promises with the intention of later breaking them.”
- The person in trouble wants his promise to be believed so he can get what he needs.
- Universalize rule: Everyone may make & break promises
- Everyone breaking promises would make promises unbelievable, contradicting desire to have promise believed
- The rule is flawed. The answer is “No.”

Categorical Imperative (2nd Formulation)

Act so that you treat both yourself
and other people as ends in themselves
and never only as a means to an end.

This is usually an easier formulation to
work with than the first formulation of the
Categorical Imperative.

2nd Formulation of Categorical Imperative



Plagiarism Scenario

- Carla
 - Single mother
 - Works full time
 - Takes two evening courses/semester
- History class
 - Requires more work than normal
 - Carla earning an “A” on all work so far
 - Carla doesn’t have time to write final report
- Carla purchases report and submits it as her own work

Kantian Evaluation (1st Formulation)

- Carla wants credit for plagiarized report
- Rule: “You may claim credit for work performed by someone else”
- If rule universalized, reports would no longer be credible indicator’s of student’s knowledge, and professors would not give credit for reports
- Proposed moral rule is self-defeating
- It is wrong for Carla to turn in a purchased report

Kantian Evaluation (2nd Formulation)

- Carla submitted another person's work as her own
- She attempted to deceive professor
- She treated professor as a means to an end
 - End: passing the course
 - Means: professor issues grade
- What Carla did was wrong

Case for Kantianism

- Rational
- Produces universal moral guidelines
- Treats all persons as moral equals
- Workable ethical theory

Perfect and Imperfect Duties

- Perfect duty: duty obliged to fulfill without exception
 - Example: Telling the truth
- Imperfect duty: duty obliged to fulfill in general but not in every instance
 - Example: Helping others

Case Against Kantianism

- Sometimes no rule adequately characterizes an action
- Sometimes there is no way to resolve a conflict between rules
 - In a conflict between a perfect duty and an imperfect duty, perfect duty prevails
 - In a conflict between two perfect duties, no solution
- Kantianism allows no exceptions to perfect duties
- Despite weaknesses, a workable ethical theory

Act Utilitarianism

(consequentialist theory)

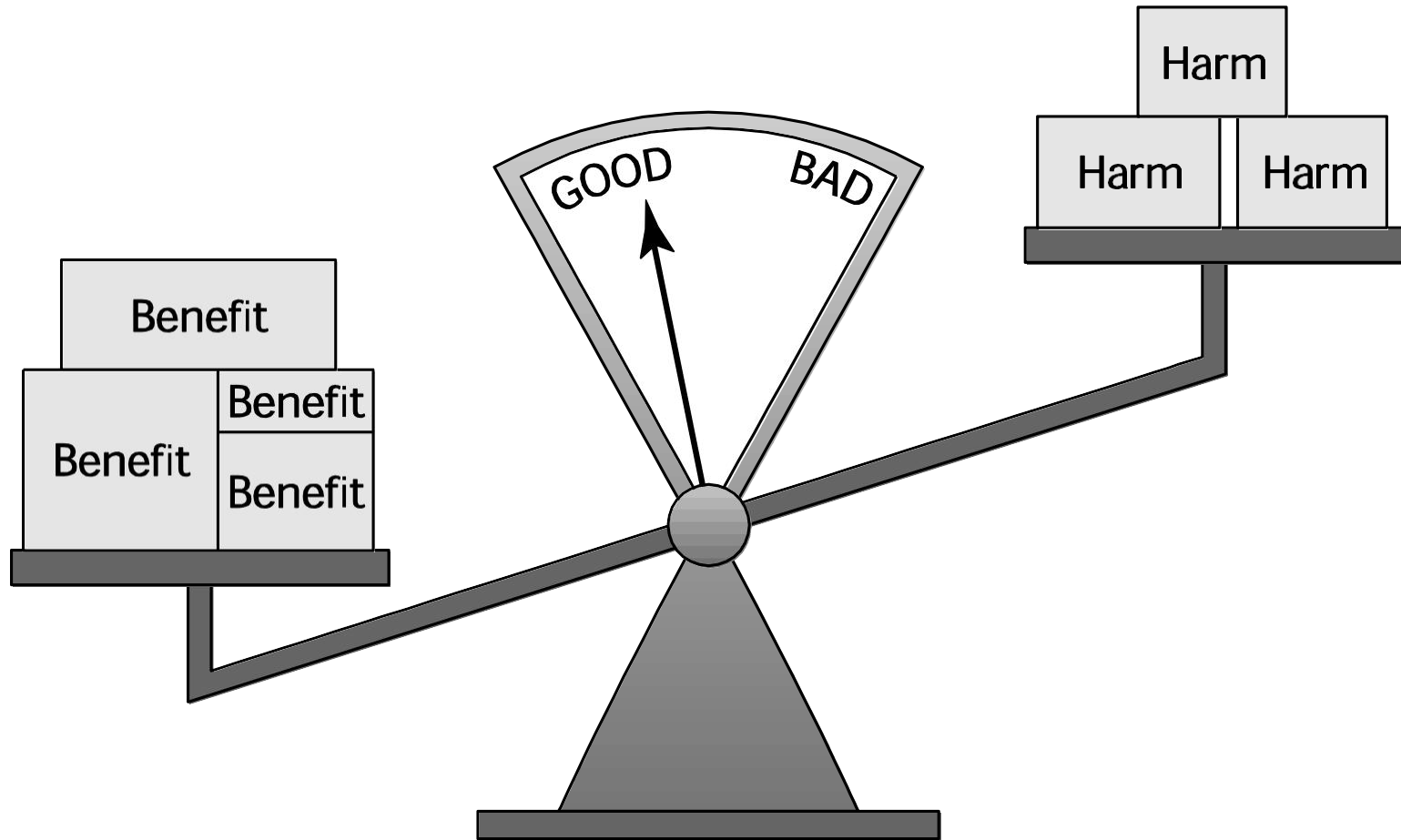
Principle of Utility

- Jeremy Bentham and John Stuart Mill
- An action is good if it benefits someone
- An action is bad if it harms someone
- Utility: tendency of an object to produce happiness or prevent unhappiness for an individual or a community
- Happiness = advantage = benefit = good = pleasure
- Unhappiness = disadvantage = cost = evil = pain

Principle of Utility (Greatest Happiness Principle)

An action is right (or wrong) to the extent that it increases (or decreases) the total happiness of the affected parties.

Principle of Utility



Act Utilitarianism

- Utilitarianism
 - Morality of an action has nothing to do with intent
 - Focuses on the consequences
 - A consequentialist theory
- Act utilitarianism
 - Add up change in happiness of all affected beings
 - $\text{Sum} > 0$, action is good
 - $\text{Sum} < 0$, action is bad

Bentham: Weighing Pleasure/Pain

- Intensity: magnitude of the experience
- Duration: how long the experience lasts
- Certainty: probability it will actually happen
- Propinquity: how close the experience is in space and time
- Fecundity: its ability to produce more experiences of the same kind
- Purity: extent to which pleasure is not diluted by pain or vice versa
- Extent: number of people affected

Highway Routing Scenario

- State may replace a curvy stretch of highway
- New highway segment 1 mile shorter
- 150 houses would have to be removed
- Some wildlife habitat would be destroyed

Evaluation

- Costs
 - \$20 million to compensate homeowners
 - \$10 million to construct new highway
 - Lost wildlife habitat worth \$1 million
- Benefits
 - \$39 million savings in automobile driving costs
- Conclusion
 - Benefits exceed costs
 - Building highway is a good action

Case for Act Utilitarianism

- Focuses on happiness
- Down-to-earth (practical)
- Comprehensive
- Workable ethical theory

Case Against Act Utilitarianism

- Unclear whom to include in calculations
- Too much work
- Ignores our innate sense of duty
- Susceptible to the problem of moral luck

Rule Utilitarianism

(Principle of Utility to moral rules)

Applying Principle of Utility to Rules

- We ought to adopt moral rules which, if followed by *everyone*, will lead to the greatest increase in total happiness
- Act utilitarianism applies Principle of Utility to individual moral actions
- Rule utilitarianism applies Principle of Utility to moral rules

Anti-Worm Scenario

- August 2003: Blaster worm infected thousands of Windows computers
- Soon after, Nachi worm appeared
 - Took control of vulnerable computer
 - Located and destroyed copies of Blaster
 - Downloaded software patch to fix security problem
 - Used computer as launching pad to try to “infect” other vulnerable PCs

Evaluation using Rule Utilitarianism

- Proposed rule: If I can write a helpful worm that removes a harmful worm from infected computers and shields them from future attacks, I should do so
- Who would benefit
 - People who do not keep their systems updated
- Who would be harmed
 - People who use networks
 - People who's computers are invaded by buggy anti-worms
 - System administrators
- Conclusion: Harm outweighs benefits. Releasing anti-worm is wrong.

Case for Rule Utilitarianism

- Compared to act utilitarianism, it is easier to perform the utilitarian calculus.
- Not every moral decision requires performing utilitarian calculus.
- Moral rules survive exceptional situations
- Avoids the problem of moral luck
- Workable ethical theory

Case Against Utilitarianism in General

- All consequences must be measured on a single scale.
 - All units must be the same in order to do the sum
 - In certain circumstances utilitarians must quantify the value of a human life
- Utilitarianism ignores the problem of an unjust distribution of good consequences.
 - Utilitarianism does **not** mean “the greatest good of the greatest number”
 - That requires a principle of justice
 - What happens when a conflict arises between the Principle of Utility and a principle of justice?
- Despite weaknesses, both act utilitarianism and rule utilitarianism are workable ethical theories

Social Contract Theory

Basis of Social Contract Theory

- Thomas Hobbes
 - “State of nature”
- We implicitly accept a social contract
 - Establishment of moral rules to govern relations among citizens
 - Government capable of enforcing these rules
- Jean-Jacques Rousseau
 - In ideal society, no one above rules
 - That prevents society from enacting bad rules

James Rachels's Definition

“Morality consists in the set of rules, governing how people are to treat one another, that rational people will agree to accept, for their mutual benefit, on the condition that others follow those rules as well.”

Kinds of Rights

- **Negative right:** A right that another can guarantee by leaving you alone
- **Positive right:** A right obligating others to do something on your behalf
- **Absolute right:** A right guaranteed without exception
- **Limited right:** A right that may be restricted based on the circumstances

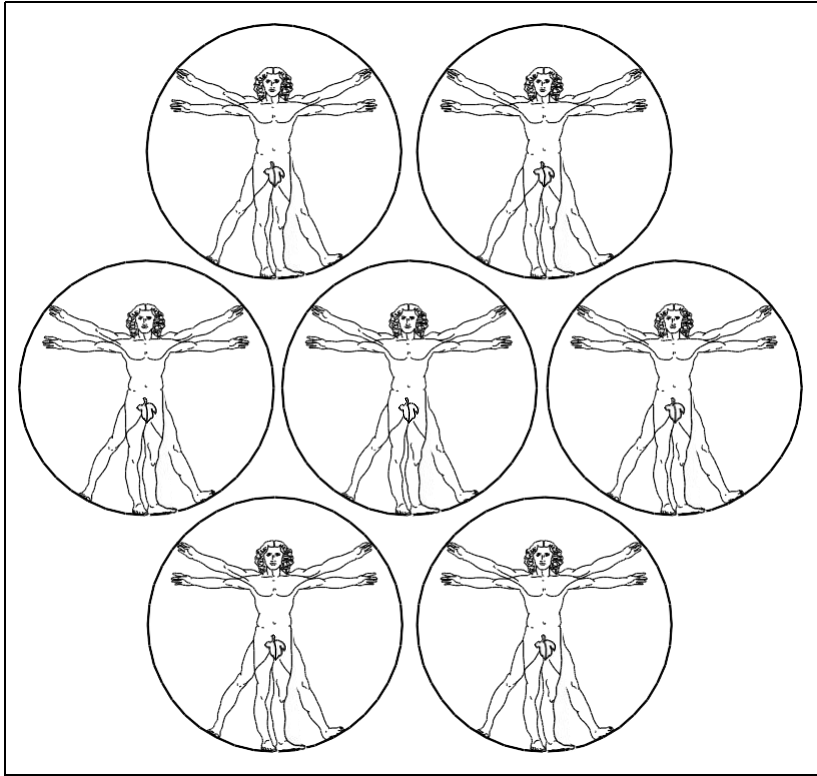
Correlation between Types of Rights

- Positive rights tend to be more limited
- Negative rights tends to be more absolute

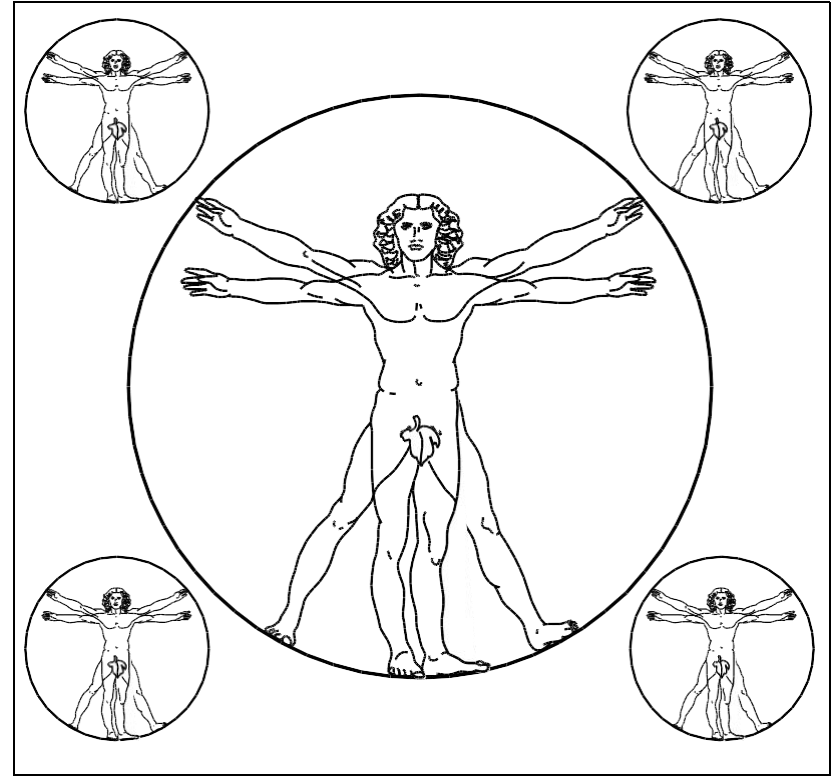
John Rawls's Principles of Justice

- Each person may claim a “fully adequate” number of basic rights and liberties, so long as these claims are consistent with everyone else having a claim to the same rights and liberties
- Any social and economic inequalities must
 - Be associated with positions that everyone has a fair and equal opportunity to achieve
 - Be to the greatest benefit of the least-advantaged members of society (the difference principle)

Rawls's First Principle of Justice

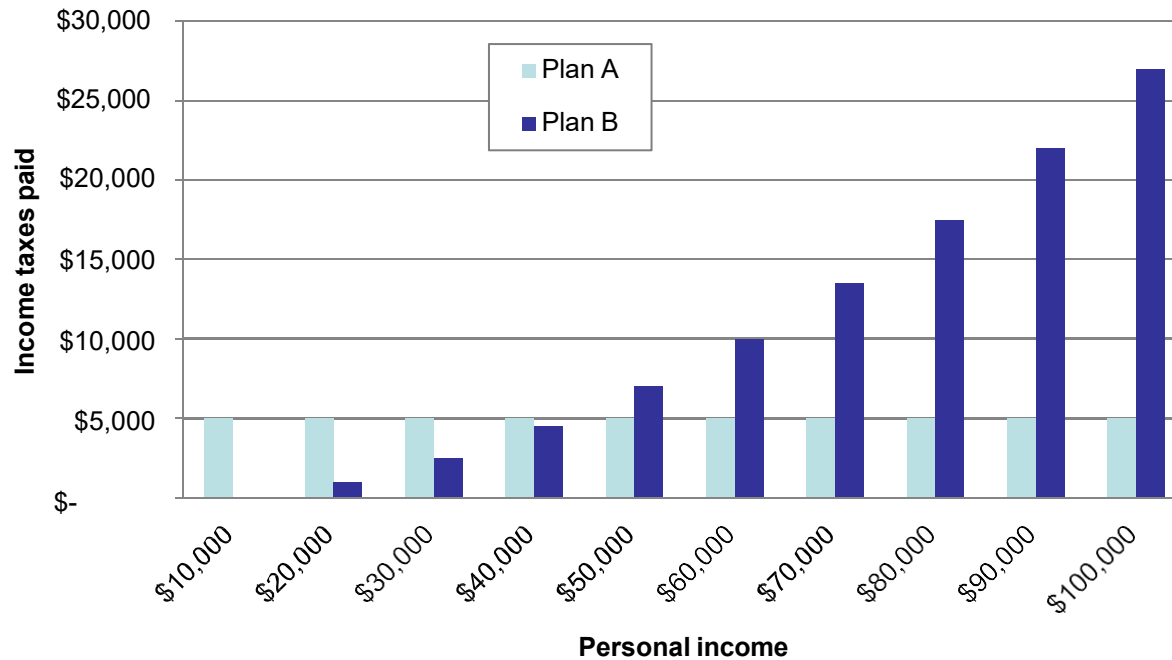


Just



Unjust

Rawls's Difference Principle



DVD Rental Scenario

- Bill owns chain of DVD rental stores
- Collects information about rentals from customers
- Constructs profiles of customers
- Sells profiles to direct marketing firms
- Some customers happy to receive more mail order catalogs; others unhappy at increase in “junk mail”

Evaluation (Social Contract Theory)

- Consider rights of Bill, customers, and mail order companies.
- Does customer have right to expect name, address to be kept confidential?
- If customer rents DVD from bill, who owns information about transaction?
- If Bill and customer have equal rights to information, Bill did nothing wrong to sell information.
- If customers have right to expect name and address or transaction to be confidential without giving permission, then Bill was wrong to sell information without asking for permission.

Case for Social Contract Theory

- Framed in language of rights
- Explains why people act in self-interest without common agreement
- Provides clear analysis of certain citizen/government problems
- Workable ethical theory

Case Against Social Contract Theory

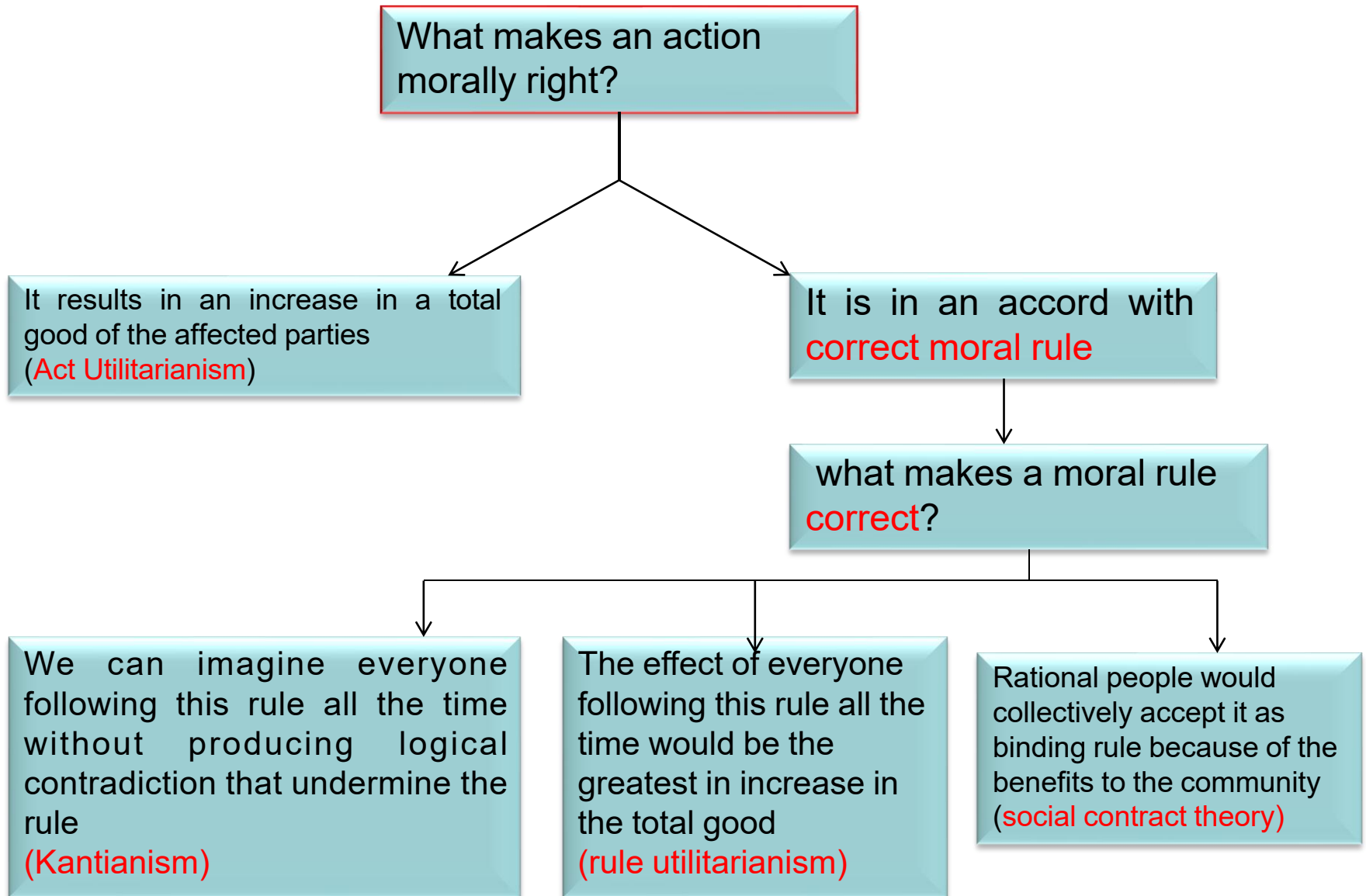
- No one signed contract
- Some actions have multiple characterizations
- Conflicting rights problem
- May unjustly treat people who cannot uphold contract
- Despite weaknesses, a workable theory

Comparing Workable Ethical Theories

Objectivism vs. Relativism

- Objectivism: Morality has an existence outside the human mind
- Relativism: Morality is a human invention
- Kantianism, utilitarianism, and social contract theory examples of objectivism

Comparing Workable Ethical Theories



Morality of Breaking the Law

Social Contract Theory Perspective

- Everyone in society bears certain *burdens* in order to receive certain *benefits*
- Legal system supposed to guarantee people's *rights are protected*
- Everything else being equal, we should be *law-abiding*
- Should only break law (*disobedience*) if compelled to follow a *higher-order moral obligation*
- *It is called morality of breaking the law*

Social Contract: A Prima Facie Obligation to Obey the Law



Prima Facie : Latin expression “prima facie “ means

- ☐ “at first sight”,
- ☐ “at first view”, or
- ☐ “based on first impression.”

Kantian Perspective

- Everyone wants to be treated justly
- Imagine rule: “I may break a law I believe to be unjust”
- If everyone acted according to this rule, then laws would be *subverted*
- **Contradiction**: Cannot both wish to be treated justly and allow laws to be subverted

Rule Utilitarian Perspective

- What would be consequences of people ignoring laws they felt to be unjust?
- **Beneficial consequence:** Happiness of people who are doing what they please
- **Harmful consequences:** Harm to people directly affected by lawless actions, general loss of respect for laws, increased burden on criminal justice system
- Harms greater than benefits

Summary

Insights Offered by Various Theories

- Kantianism: Interactions with other people should *respect* them as *rational beings*
- Utilitarians: You should consider the *consequences* of an action before deciding whether it's right or wrong
- **Social contract theory:** We should promote *collective rights*, such as the rights to *life, liberty, and property*

Mixing Theories

- You can consider *duties and rights and consequences* when making moral decisions
- But what will you do when you can't respect *rights absolutely* and still maximize the total ***beneficial consequences?***
- Contemplation of what it means to be a person of good character leads to a discussion of *virtue ethics*

Thanks